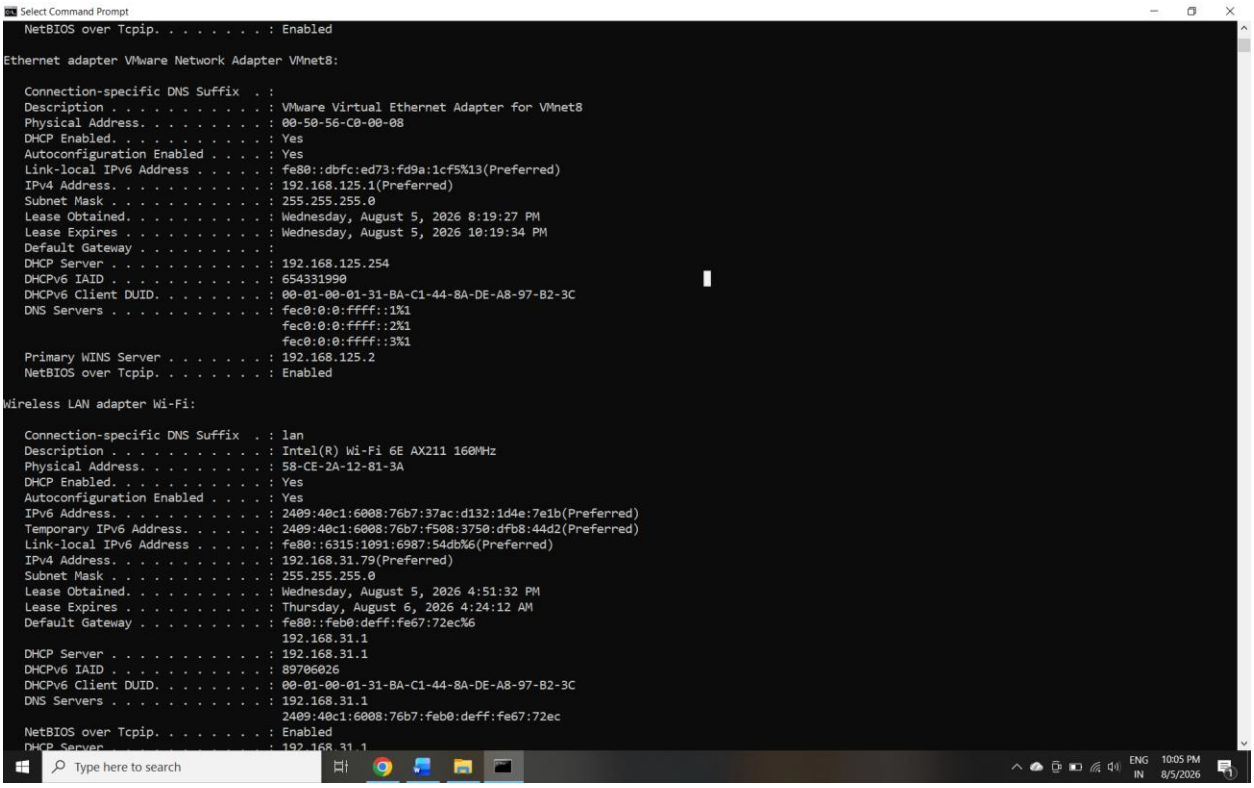
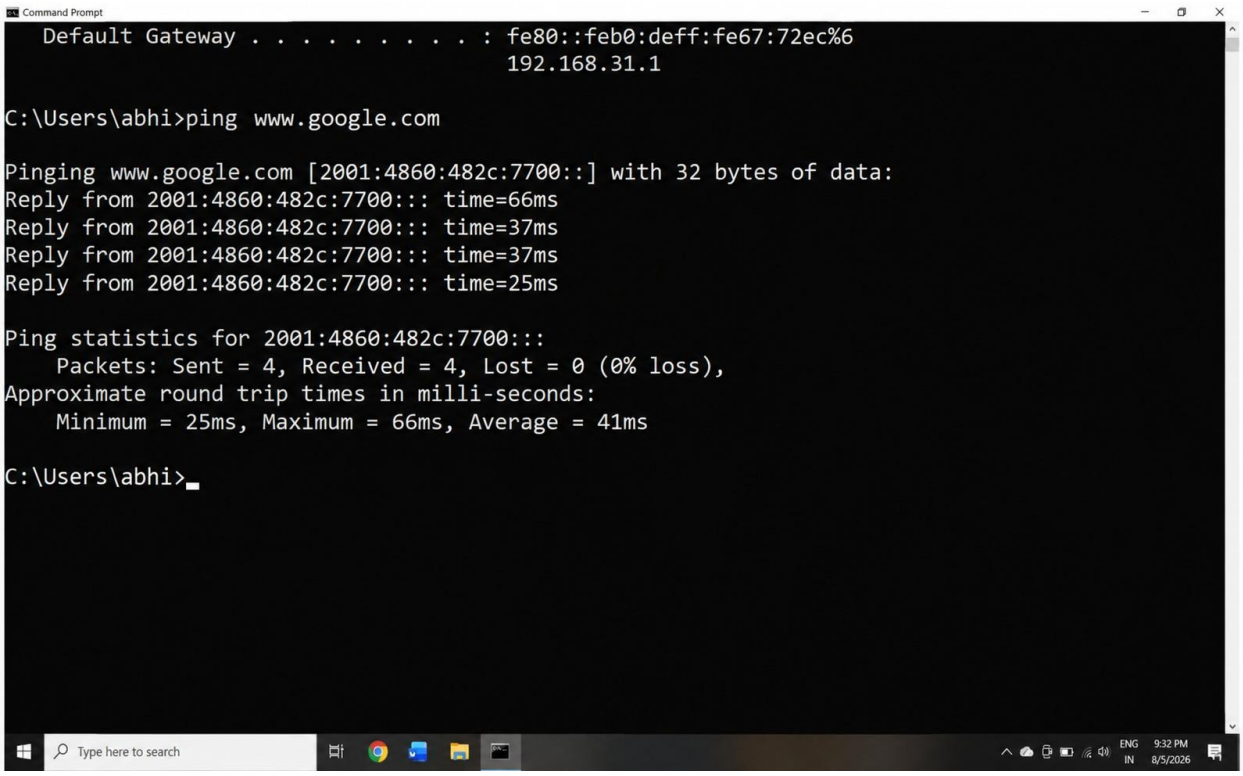
1.Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

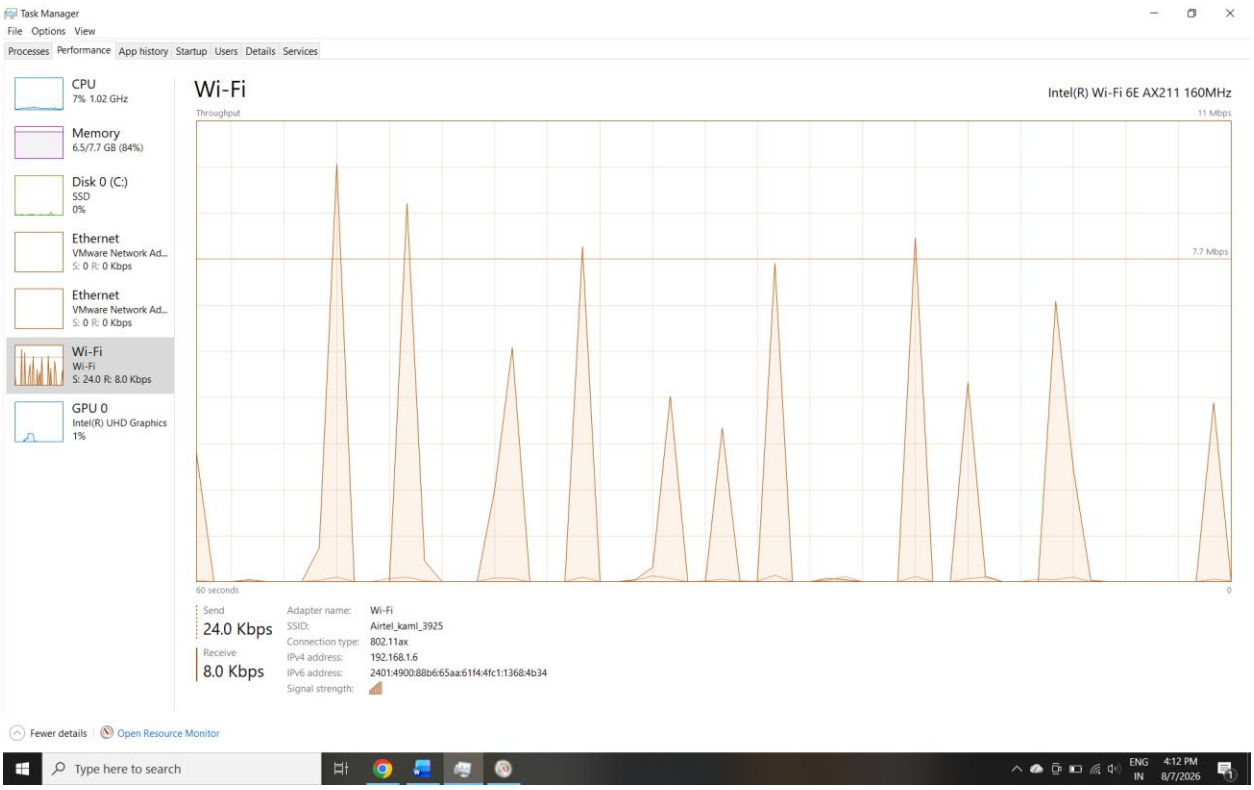


2.Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

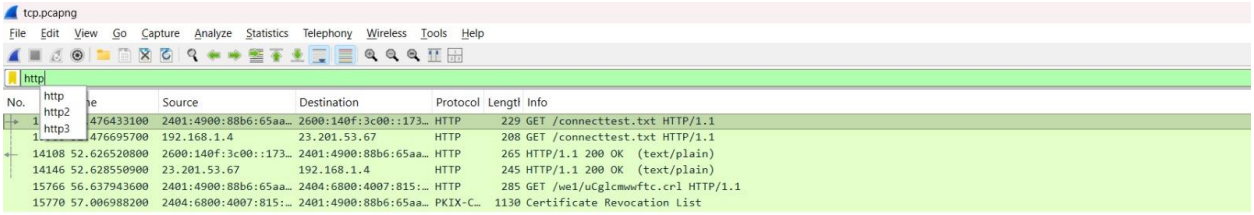


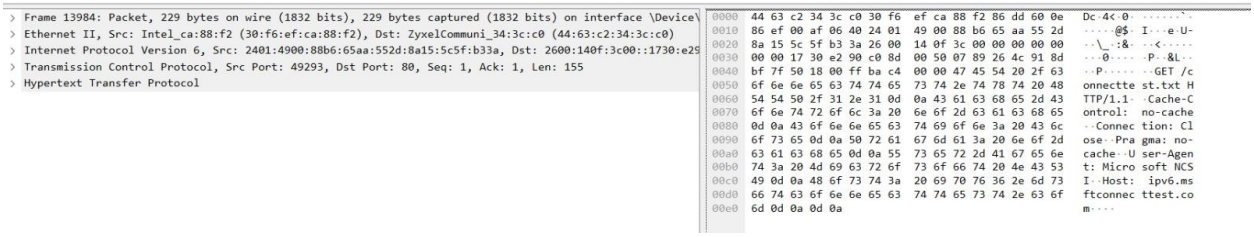
When the network was disconnected, the ping command failed because the computer had no connection to the internet and could not resolve or reach Google's server. After reconnecting the network, ping received replies successfully, confirming that internet connectivity was restored. The difference between both results helps identify whether the problem is due to a local network connection issue.

3.Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.



4.Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.





5.Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Restart Router

Turn off the router for a few seconds and turn it on again.

Check Internet Speed

Check your internet speed and ping using a speed test.

Use Ethernet Cable

Connect your PC or console directly to the router for a stable connection.

Disconnect Extra Devices

Remove devices that are using too much internet.

Move Router Location

Keep the router near your gaming device and away from obstacles.

Update Software

Update router firmware and device network drivers.